Classification of Signals:

① Continuous-time and discrete-time signals:

Basis of classification: how they are defined as a function of time

→ a signal $x(t)$ is said to be continuous-time signal if it is defined for all time t.

→ Continuous-time signals arise naturally when a physical waveform such as an accoustic wave or light wave is converted into an electrical signal.

e.g. microphone, which converts variation in sound pressure into corresponding variations in voltage and current, and the photocell, which does the same for variations in light intensity.

→ Discrete-time signals are defined only at discrete instants of time. The independent variable has discrete value only, which are usually uniformly spaced. A discrete signal is often derived from a continuous-time signal by sampling it at a uniform rate. Let T_s denote the sampling period and n denote an integer that may assume +ve & −ve values. Then sampling a continuous-time signal $x(t)$ at time $t = nT_s$ yields a sample with the value $x(nT_s)$.

$$\boxed{x[n] = x(nT_s)}$$

Discrete-time signal is represented by the sequence of numbers $\ldots, x[-2], x[-1], x[0], x[1], x[2], \ldots$ which can take a continuum of values. This is also called time series, written as $\{x[n], n = 0, \pm 1, \ldots\}$, i.e., $x[n]$.

Music to a CD: 44100 Hz — Sampling rate
16 bit — Bit-depth
} But while recording it is generally higher than this.

To do:

1. Record voice using audacity at two different frequencies.
2. Hear them.
3. See the change in size of matrix.

2 Even and odd signals:

A continuous signal $x(t)$ is said to be an even signal if

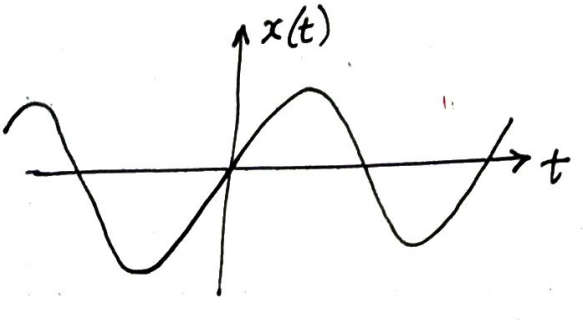
$$x(-t) = x(t) \quad \forall\ t \in \mathbb{R}$$

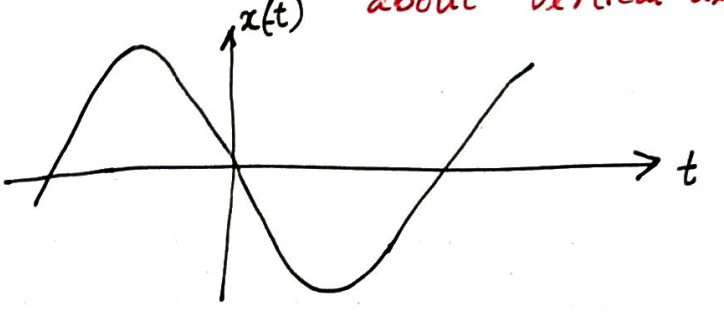
← symmetric about vertical axis

The signal is said to be an odd signal if

$$x(-t) = -x(t) \quad \forall\ t \in \mathbb{R}$$

← antisymmetric about vertical axis





e.g.

Consider the signal

$$x(t) = \begin{cases} \sin\left(\frac{\pi t}{T}\right), & -T \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

Is the signal $x(t)$ an even or an odd function of time t?

<u>Solution:</u>

$$x(-t) = \begin{cases} \sin\left(\frac{-\pi t}{T}\right), & -T \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

$$= \begin{cases} -\sin\left(\frac{\pi t}{T}\right), & -T \leq t, \leq T \\ 0, & \text{otherwise} \end{cases}$$

$$= -x(t) \quad \forall \; t \in \mathbb{R}$$

$\rightarrow$ $x(t)$ is an odd signal.

Every arbitrary signal can be decomposed into even-odd part.

$$x(t) = x_e(t) + x_o(t) \longrightarrow \text{(I)}$$

$$\Rightarrow x(-t) = x_e(-t) + x_o(-t)$$

$$\Rightarrow x(-t) = x_e(+t) - x_o(t) \longrightarrow \text{(II)}$$

$$\text{(I)} + \text{(II)} \Rightarrow x(t) + x(-t) = 2x_e(t) \Rightarrow \boxed{x_e(t) = \frac{1}{2}\left(x(t) + x(-t)\right)}$$

$$\text{(I)} - \text{(II)} \Rightarrow x(t) - x(-t) = 2x_o(t) \Rightarrow \boxed{x_o(t) = \frac{1}{2}\left(x(t) - x(-t)\right)}$$

e.g. $x(t) = e^{-2t}\cos t$

$$x(-t) = e^{2t}\cos(-t) = e^{2t}\cos t$$

$$\therefore\ x_e(t) = \frac{1}{2}\left(e^{-2t}\cos t + e^{2t}\cos t\right) = \frac{1}{2}\cos t\left(e^{-2t}+e^{2t}\right)$$

$$= \cosh(2t)\cos t$$

$$x_o(t) = \frac{1}{2}\left(e^{-2t}\cos(t) - e^{2t}\cos t\right) = \frac{1}{2}\cos t\left(e^{-2t}-e^{2t}\right)$$

$$= -\sinh(2t)\cos t.$$

<u>For complex valued signal</u>:

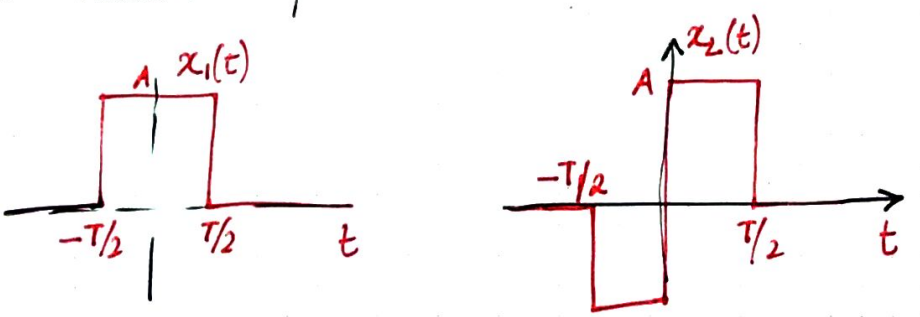
A complex valued signal $x(t)$ is said to be conjugate symmetric if $x(-t) = x^*(t)$.

Let $x(t) = a(t) + j\,b(t)$

$$\left.\begin{aligned} x(-t) &= a(-t) + j\,b(-t) \\ x^*(t) &= a(t) - j\,b(t) \end{aligned}\right\} \quad \begin{aligned} a(t) &= a(-t) \quad \text{even} \\ b(t) &= -b(-t) \quad \text{odd} \end{aligned}$$

Similar remark for the discrete case also.

e.g.



③ Periodic signals and nonperiodic signals:

A periodic signal $x(t)$ is a function of time that satisfies

$$x(t) = x(t+T) \quad \text{for all } t \in \mathbb{R}, \text{ where } T \in \mathbb{R}. \quad \rightarrow ⓐ$$

Clearly, if the above condition is satisfied then

$$x(t) = x(t+T) = x(t+2T) = \cdots = x(t+kT) \ \forall \ t \in \mathbb{R}.$$

The smallest value of T that satisfies eq. ⓐ is called the fundamental period of $x(t)$.

$1/T$ is called the fundamental frequency.

The angular frequency is $\omega = 2\pi/T$.

If $\nexists \ T \in \mathbb{R}$ that satisfies equation ⓐ then the signal is called aperiodic / nonperiodic.

<u>For discrete case:</u>

A discrete-time signal $x[n]$ is said to be periodic if

$$x[n] = x[n+N] \quad \text{for integer } n, \quad \rightarrow ⓑ$$

where N is a <u>positive</u> integer.

The smallest integer N for which ⓑ is satisfied is called the fundamental period of the discrete-time signal $x[n]$. The fundamental angular frequency is $\Omega = 2\pi/N$.

4. Deterministic signals and random signals:

A deterministic signal is a signal about which there is no uncertainity with respect to its value at any time.

By contrast, a random signal is a signal about which there is uncertainty before it occurs. Such a signal can be viewed as an ensemble of signals with each signal in the ensemble having a different waveform. Each signal within the ensemble had certain probability of occurrence.

e.g. The electrical noise generated in the amplifier of a radio or a television receiver is an example of random noise.

Another example is the signal received in a radio communication system. The signal consists of an information-bearing components, an interference component, and unavoidable electrical noise generated at the front end of the receiver.

Information bearing component — voice signal

Interference component — spurious EM waves produced by other components.

The net result is a random signal.